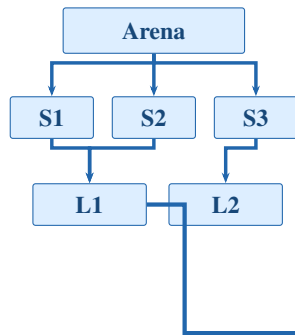


Arena / Streams / Layers

HEAS UNIFIED PATH (with contract):



metrics_episode() Contract

DIVERGENT PATH (without contract):

Interface

```
def metrics_episode(  
    obs, actions,  
    rewards  
) → metrics_dict
```

metrics_episode()

✓ NO DIVERGENCE

Pipeline Stages

Optimizer

Tournament

Inference

mean_biomass

q75_biomass

entropy

Example — Policy A (Rank inconsistency)

Divergent: Stage₁ uses mean_biomass = 50.5 → Stage₂ uses q75_biomass = 57.5 ↑ (different metric ⇒ rank #3 → rank #7)

HEAS: All stages call metrics_episode() → mean_biomass = 50.5 everywhere (rank #3 stays #3, consistent)

Contract: metrics_episode() registered once ⇒ all pipeline stages read identical metric values ⇒ no silent aggregation divergence.